Lektion 5: Sundhedsomkostninger

Kim Rose Olsen

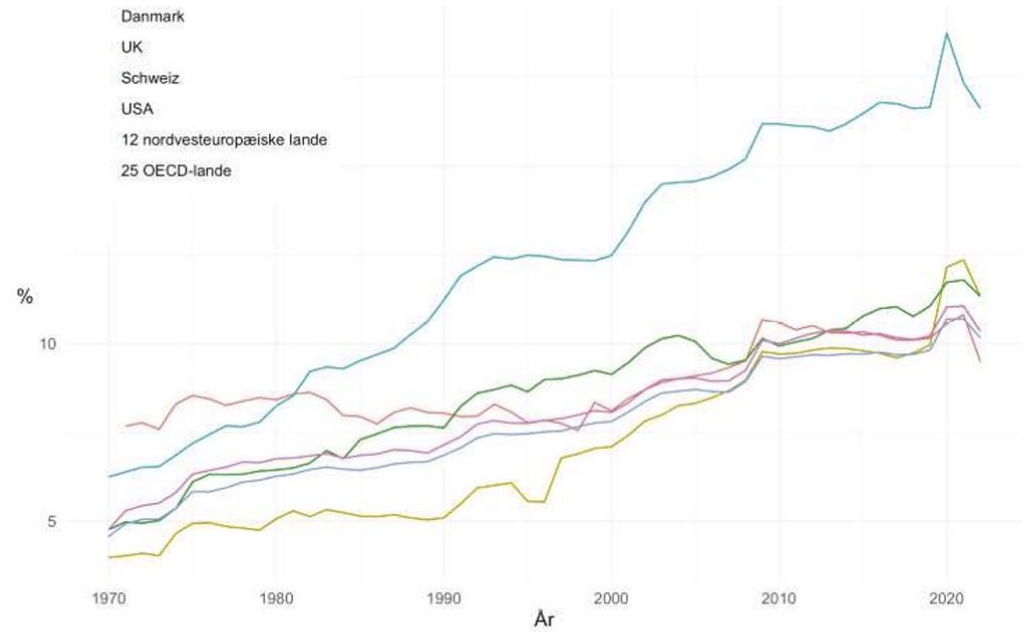
SDU
Danish Centre for Health Economics
- DaCHE

Indhold i lektion 3

- Introduktion til sundhedsomkostninger og omkostningsregistre:
    - Hospital: Landspatientsregistret og DRG
    - Primær sektor: Sygesikringsregistret
    - Lægemidler: Lægemiddelregistret

Udvikling i sundhedsudgifterne

Figur 2: Totale udgifter til sundhed som procent af BNP i udvalgte lande, 12 nordvesteuropæiske lande og 25 OECD-lande

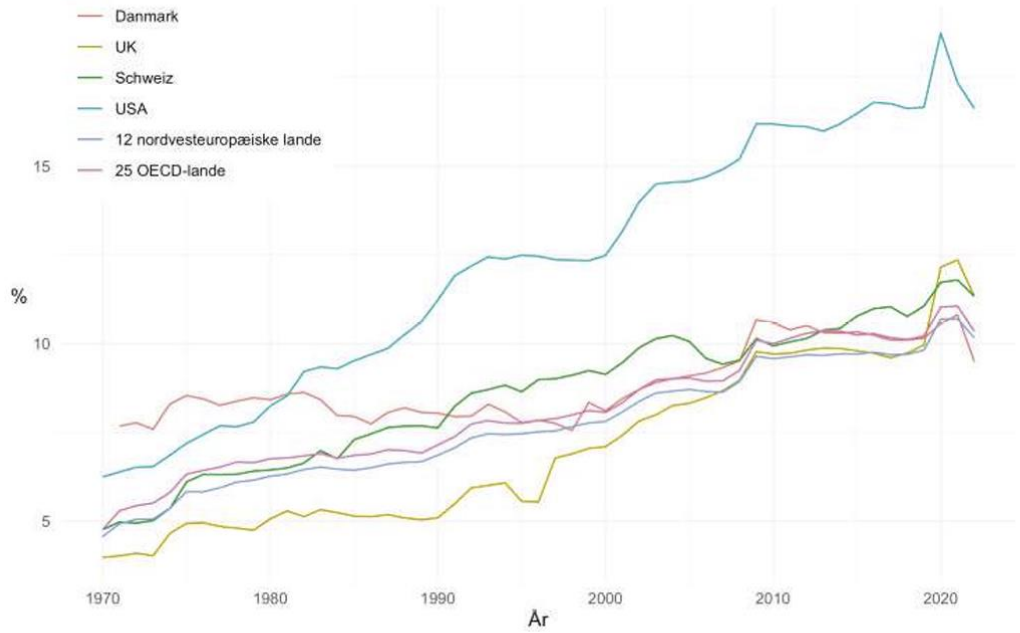


Kilde: OECD.stat (2024), Health expenditure and financing.

Kallestrup-Lamp, M., Marin, A. O. K., & Søgaard, J. (2024).
Befolkningsaldring og sundhedsudgifter i Danmark. Økonomi & Politik, 97(4), 10-25. https://tidsskrift.dk/okonomi-og-politik/article/view/151051

Udvikling i sundhedsudgifterne

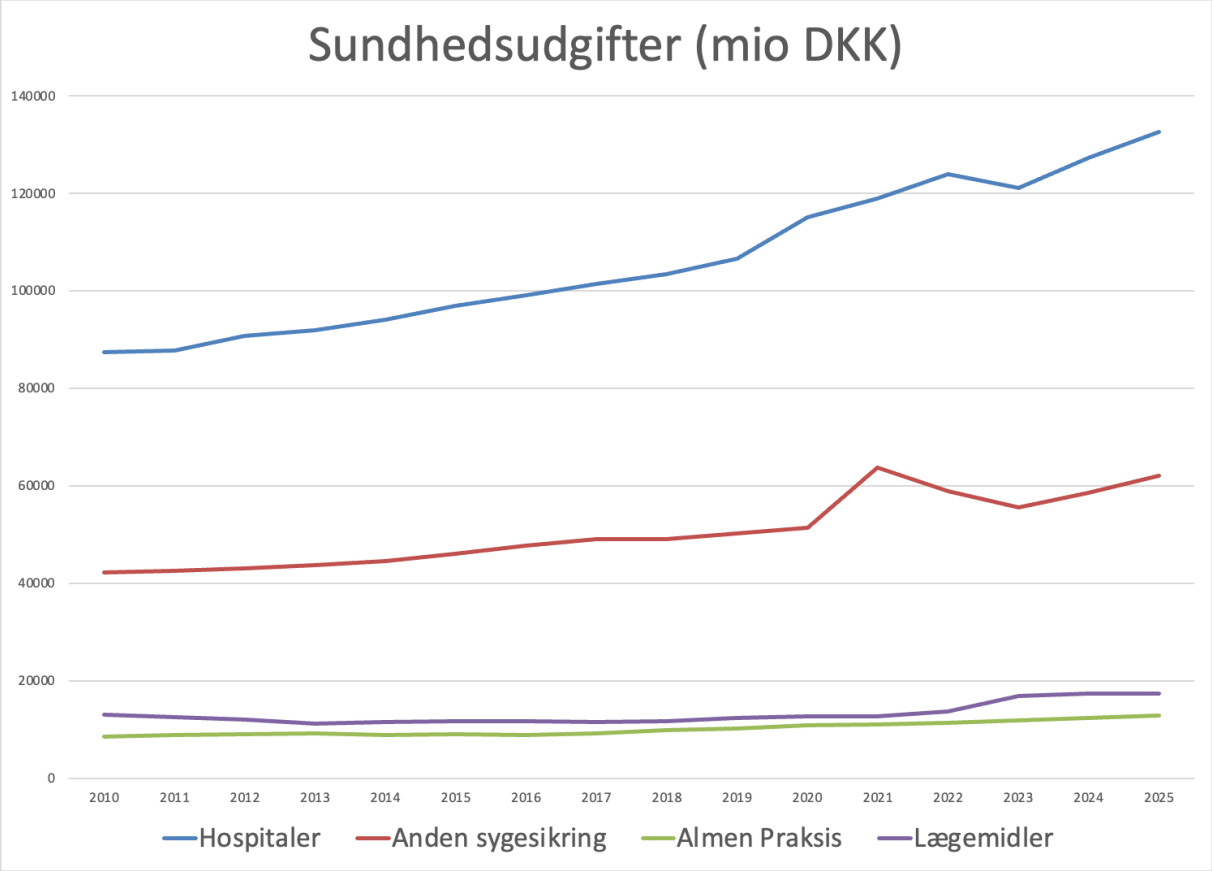
Figur 2: Totale udgifter til sundhed som procent af BNP i udvalgte lande, 12 nordvesteuropæiske lande og 25 OECD-lande



Kilde: OECD.stat (2024), Health expenditure and financing.

Kallestrup-Lamp, M., Marin, A. O. K., & Søgaard, J. (2024).
Befolkningsaldring og sundhedsudgifter i Danmark. Økonomi & Politik, 97(4), 10-25. https://tidsskrift.dk/okonomi-og-politik/article/view/151051

Fordeling mellem sektorer



Kilde: statistikbanken.dk

Typiske “omkostnings studier”



Journal of Psychiatric Research 144 (2021) 331–337

Contents lists available at ScienceDirect

Journal of Psychiatric Research

journal homepage: www.elsevier.com/locate/jpsychires

The impact of patient-controlled hospital admissions among patients with severe mental disorders on health care cost: A nationwide register-based cohort study using quasi-experimental design

Louise Schubert Paaske [a,*], Liza Sopina [a], Kim Rose Olsen [a], Christoffer Torgaard Thomsen [b], Michael Eriksen Benros [b,c], Merete Nordentoft [b], Lene Halling Hastrup [a,d]

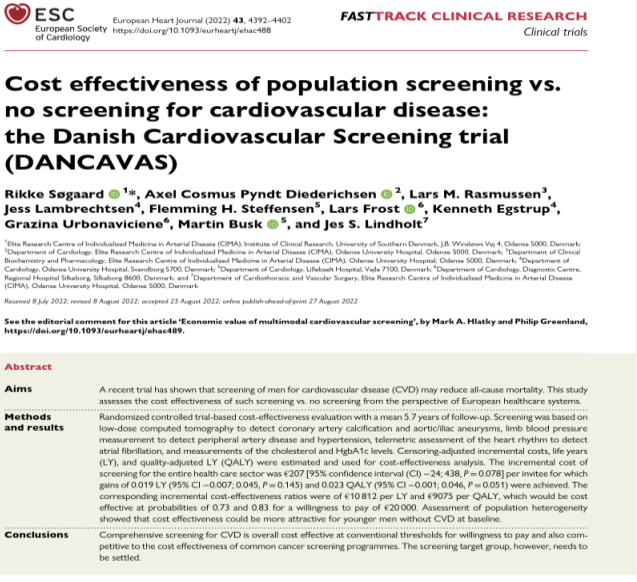
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ARTICLE INFO

Keywords:
Mental health services
Patient admission
Psychiatry health care costs

ABSTRACT

To assess whether patient-controlled admissions (PCA) to psychiatric hospital wards have an effect on t
health care costs in the short term and longer term compared with patients receiving treatment as usual (TA
Based on Danish register data and using coarsened exact matching, patients who signed a contract for PCA dur
2013–2016 were matched on observables with a control group of TAU patients identified from psychia
registers. As primary outcome patients' expenditure was explored in 6-monthly intervals over the first
months, and as secondary outcome over the 12–24 month period after index month. To control for unobserva
selection a quasi-experimental difference-in-difference design was applied on the matched population using
principle of intention to treat. 414 PCA patients (out of 423 available) were successfully matched with 7
never exposed TAU patients (out of 67,331 available). Total healthcare costs for PCA patients were €8887 (9
CI 708–17,067) higher per month than unexposed TAU patients in the first six months after index. Mont
psychiatric costs were €8922 (95% CI 708–17,067) higher for PCA patients than for TAU patients. These
ferences persisted, in the next three six-monthly periods, albeit were not statistically significant, in the sec
six-monthly period after index. In conclusion, driven by the psychiatric costs, implementing PCA increased
total healthcare costs in the short term compared with TAU. More research of the impact on health care costs
broader and long-term perspective is still needed.



ESC European Society of Cardiology

European Heart Journal (2022) 43, 4392–4402
https://doi.org/10.1093/eurheartj/ehac488

FASTTRACK CLINICAL RESEARCH
Clinical trials

Cost effectiveness of population screening vs. no screening for cardiovascular disease: the Danish Cardiovascular Screening trial (DANCAVAS)

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[1]Elite Research Centre of Individualized Medicine in Arterial Disease (CIMA), Institute of Clinical Research, University of Southern Denmark, J.B. Winslows Vej 4, Odense 5000, Denmark; [2]Department of Cardiology, Elite Research Centre of Individualized Medicine in Arterial Disease (CIMA), Odense University Hospital, Odense 5000, Denmark; [3]Department of Clinical Biochemistry and Pharmacology, Elite Research Centre of Individualized Medicine in Arterial Disease (CIMA), Odense University Hospital, Odense 5000, Denmark; [4]Department of Cardiology, Odense University Hospital, Svendborg 5700, Denmark; [5]Department of Cardiology, Lillebaelt Hospital, Vejle 7100, Denmark; [6]Department of Cardiology, Diagnostic Centre, Regional Hospital Silkeborg, Silkeborg 8600, Denmark; and [7]Department of Cardiothoracic and Vascular Surgery, Elite Research Centre of Individualized Medicine in Arterial Disease (CIMA), Odense University Hospital, Odense 5000, Denmark

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See the editorial comment for this article 'Economic value of multimodal cardiovascular screening', by Mark A. Hlatky and Philip Greenland, https://doi.org/10.1093/eurheartj/ehac489.

Abstract

Aims A recent trial has shown that screening of men for cardiovascular disease (CVD) may reduce all-cause mortality. This study assesses the cost effectiveness of such screening vs. no screening from the perspective of European healthcare systems.

Methods and results Randomized controlled trial-based cost-effectiveness evaluation with a mean 5.7 years of follow-up. Screening was based on low-dose computed tomography to detect coronary artery calcification and aortic/iliac aneurysms, limb blood pressure measurement to detect peripheral artery disease and hypertension, telemetric assessment of the heart rhythm to detect atrial fibrillation, and measurements of the cholesterol and HgbA1c levels. Censoring-adjusted incremental costs, life years (LY), and quality-adjusted LY (QALY) were estimated and used for cost-effectiveness analysis. The incremental cost of screening for the entire health care sector was €207 [95% confidence interval (CI) −24; 438, $P = 0.078$] per invitee for which gains of 0.019 LY (95% CI −0.007; 0.045, $P = 0.145$) and 0.023 QALY (95% CI −0.001; 0.046, $P = 0.051$) were achieved. The corresponding incremental cost-effectiveness ratios were of €10 812 per LY and €9075 per QALY, which would be cost effective at probabilities of 0.73 and 0.83 for a willingness to pay of €20 000. Assessment of population heterogeneity showed that cost effectiveness could be more attractive for younger men without CVD at baseline.

Conclusions Comprehensive screening for CVD is overall cost effective at conventional thresholds for willingness to pay and also competitive to the cost effectiveness of common cancer screening programmes. The screening target group, however, needs to be settled.



Health policy 125 (2021) 760–767

Contents lists available at ScienceDirect

Health policy

journal homepage: www.elsevier.com/locate/healthpol

Cost and quality impacts of treatment setting for type 2 diabetes patients with moderate disease severity: Hospital- vs. GP-based monitoring

Ryan Pulleyblank*, Mauro Laudicella, Kim Rose Olsen

Danish Centre for Heath Economics (DaCHE), Department of Public Health, University of Southern Denmark

ARTICLE INFO

Article history:
Received 29 May 2020
Revised 15 March 2021
Accepted 21 March 2021

Keywords:
Type 2 diabetes
Disease management
Cost
Quality of care
Administrative data

ABSTRACT

Objectives: This study investigates cost and quality implications of moving regular monitoring of patients with moderate severity type 2 diabetes (T2D) away from specialized hospital clinics into general practice (GP).

Methods: A 2016 cross-section of patients with moderate disease severity T2D were algorithmically identified using Danish administrative databases. 152,632 GP- and 21,359 hospital-monitored patients with T2D were identified. Total annual healthcare cost is decomposed into GP, medication, nonhospital-specialist, hospital outpatient and inpatient costs. Hospitalizations are used to proxy for quality of care. Cost and quality impacts of treatment setting are assessed using an instrumental variable (IV) analysis. A wide range of patient confounders are used to reduce selection bias, with the difference in patients' travel-time between nearest specialist outpatient clinic and GP used as an instrument to control for remaining endogeneity of treatment setting. Two-part models are used for zero-inflated outcomes.

Results: Average total annual healthcare cost were 14,056 DKK greater amongst hospital-monitored patients. IV analysis accounting for endogeneity of treatment setting indicates hospital-based monitoring is causally linked to higher total annual healthcare costs. The estimated local average treatment effect of hospital-based monitoring on total annual healthcare costs was higher (117.2K, 95% CI: 23.3K, 211.1K) than GP-based monitoring. No difference of treatment settings' quality based on evidence of hospitalizations was found.

Conclusion: For patients with moderate disease severity T2D, IV analysis accounting for treatment setting endogeneity bias identifies an expected efficiency improvement (average cost reduction without reduction of quality) of moving regular disease management from hospital-based setting to the GP setting.

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Data struktur

Omkostninger: $\sum_{t=1}^{t=T} \sum_{1}^{K} \; p_k \cdot q_{tk}$

t : tidsperioder

k : strata

Register	Dato	Aktivitet/Strata	Pris
Landspatient-registret	INDTIDSPUNKT_ DRGSGHF	PNR/DRG ICD_10	TOTALPRIS
Sygesikrings-registret	BEHDATO	PNR/Speciale/spec2	BRUHON
Lægemiddel-databasen	EKSD	PNR/Recept/ATC	EKSP

Hospitalsomkostninger

- Indlæggelse - akut
- Indlæggelse - ikke akut
- Ambulant - akut
- Ambulant - ikke akut

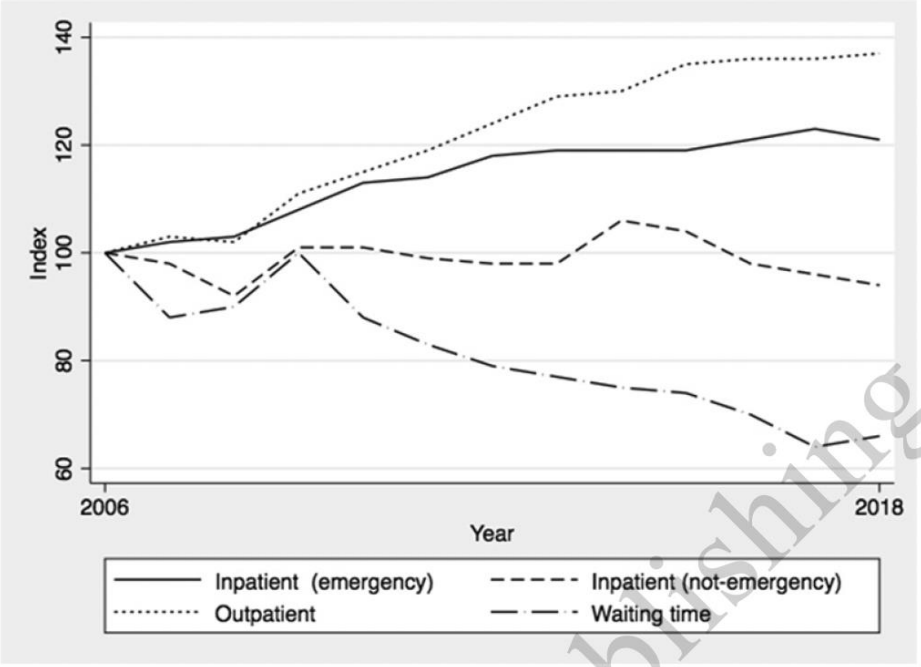


Fig. 2. Trends in Hospital Activity and Waiting Time. *Note:* Activity data are from Statistics Denmark (https://www.statistikbanken.dk) and waiting time data reflect waiting time for surgery and are from www.esundhed.dk.

Publishing

Landspatientregistret

ICD10 klassifikation

- ICD er en forkortelse for Verdenssundhedsorganisationens (WHO's) "International Statistical Classification of Diseases and Related Health Problems"

- **LINK**
- https://icd.who.int/browse10/2019/en

DRG klassifikation

- Grupperingslogik
  - Grupperne skal være klinisk meningsfulde
  - Grupperne skal være ressourcehomogene
  - Antallet af grupper skal være overskueligt

- DRG-takster.

DRG registret – vigtige variable

- DRG_SGHFORLOB : registret
- PRIORITET_DRGSGHF: akut/ikke akut
- ADIAG_DRGSGHF: opdel i sygdomstyper
- TOTALPRIS_DRGSGHF: omkostninger
- VARIGHED_TIME_TYPE_DRGSGHF: amb eller indlæggelse

Sygesikringsregistret

- **SSSY**: Data modtages årligt fra CSC Scandihealth. Der modtages 2 datasæt: Et med oplysninger om alle sygesikringsydelser givet af læger mv, og et med oplysninger om hele befolkningens fordeling på sygesikringsgrupper. CPR og yderidentifikation kan kobles.

- **SPEC2**: SPEC2 (den to-cifrede speciale kode) refererer til en bestemt type af yder, dvs. almen læge, ørelæge, tandlæge osv. eller til andre typer af ydergrupper så som laboratorieundersøgelser mv. Opdelingen i de forskellige ydertyper bygger på de forskellige overenskomster, der er indgået mellem ydergruppen og regionerne

- Ofte indeles i to kategorier:
  - Almen Praksis
  - Andet sygesikringsaktivitet

Sygesikringsregistret – AP ydelser

HONORARTABEL

DAGTID

Overenskomst om almen praksis

1. oktober 2025 til 1. april 2026

Vær også opmærksom på din regions lokalaftaler.		Oktober 25 honorar
Basishonorar		
Basishonorar pr. gruppe 1-sikrede pr. måned		44,89
Basishonorar pr. gruppe 1-sikrede pr. år		538,70
Praksisomkostningshonorar		
Praksisomkostningshonorar pr. år		19.552,82
Plejehjems- og bostedslægehonorar		
Honorar ved 2 timers rådgivning/undervisning inkl. transport pr. måned		2.159,02
Honorar ved 3 timers rådgivning/undervisning inkl. transport pr. måned og over 30 beboere		3.238,53
§ 50 Grundydelser		
Mandag – fredag kl. 08-16		
0101	Konsultation	169,11
0120	Årsstatus (aftalt specifik forebyggende indsats)[1]	440,83
0201	Telefonkonsultation	32,06
0204	Aftalt telefonkonsultation	131,01
3201	Socialmedicinsk telefonkonsultation	77,39
0124	Lægefaglig vurdering af patienter på kommunal akutfunktion	176,94
0105	E-konsultation	51,48
0127	E-konsultation med kommunen	51,48
0125	Videokonsultation[2]	169,11
1003	Registreringsydelse til 0120, 6101 og 6201, hvis de udføres på video	0,00
6101	Samtaleterapi	464,85
0128	Konsulterende samtale til børn og familier[3]	348,99
0122	Abortstøttesamtale	464,85
0411	Sygebesøg op til 4 km[4] + evt. kørselsgodtgørelse for alle kørte km	439,54
0421	Sygebesøg fra påbegyndt 5 km indtil 8 km [4] + evt. kørselsgodtgørelse for alle kørte km	604,64

“Speciale” variablen

800101
800120

spec2

Sygesikringsregistret

- **SPECIALE**: 6-cifret variabel, som består af to delelementer. De første to cifre angiver yderens speciale (altså om det er en praktiserende læge, en tandlæge osv.), mens de sidste fire cifre angiver typen af ydelsen, der er blevet givet den pågældende person. Informationer om speciale og typen af ydelse (ydelseskode) kan ses i overenskomstens takstmapper - se www.okportalen.dk og den detaljerede beskrivelse.
- **BRUHON**: Pris for ydelsen

Hvad sker der når taksterne ændres?

Received: 3 November 2022 | Revised: 6 June 2023 | Accepted: 1 July 2023
DOI: 10.1002/hec.4736

RESEARCH ARTICLE

Health Economics WILEY

Increasing capitation in mixed remuneration schemes: Effects on service provision and process quality of care

Christian Volmar Skovsgaard[1] | Troels Kristensen[1] | Ryan Pulleyblank[1,2] | Kim Rose Olsen[1]

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Funding information
Norges Forskningsråd, Grant/Award Number: 288592; Novo Nordisk Fonden, Grant/Award Number: NNF18OC0033978; European Union's EU Framework Programme for Research and Innovation Horizon 2020, Grant/Award Number: 721402

Abstract
Many health systems apply mixed remuneration schemes for general practitioners, but little is known about the effects on service provision of changing the relative mix of fee for services and capitation. We apply difference-in-differences analyses to evaluate a reform that effectively reversed the mix between fee for services and capitation from 80/20 to 20/80 for patients with type 2 diabetes. Our results show reductions in provision of both the contact services that became capitated and in other non-capitated (still-billable) services. Reduced provision also occurred for guideline-recommended process quality services. We find that the effects are mainly driven by patients with co-morbidities and by general practitioners with high income, relatively many diabetes patients, and solo practitioners. Thus, increasing capitation in a mixed remuneration schemes appears to reduce service provision for patients with type 2 diabetes monitored in general practice with a risk of unwanted quality effects.

KEYWORDS
capitation, fee-for-service, general practice, mixed schemes, natural experiment, remuneration, type 2 diabetes

JEL CLASSIFICATION
D4, D9, I18, J3, H4, H5

Health Economics

WILEY

Health Economics

RESEARCH ARTICLE OPEN ACCESS

Do Physicians Respond to Additional Capitation Payments in Mixed Remuneration Schemes?

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Correspondence: Line Planck Kongstad (lpkongstad@health.sdu.dk)

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Funding: This study was supported by Novo Nordisk Fonden (Grant Number: NNF18OC0033978) and Norges Forskningsråd (Grant Number: 288592).

Keywords: income target hypothesis | physician incentives | physician remuneration

ABSTRACT
Mixed remuneration schemes with capitation and fee-for-service (FFS) payments hold financial incentives to add patients to the list and provide services to listed patients. However, as patients with complex needs tend to require longer consultations there is a risk of inequality in access if fees are not adjusted to patient characteristics. In this paper, we assess a natural experiment introducing additional capitation for GPs with a high share of complex patients (moderate scheme) and for GPs in certain geographical areas (intensive scheme). GPs are eligible if the complexity of their listed patients exceeds a threshold, but as the scheme is subject to a national budget constraint, some eligible general practitioners (GPs) are left without additional payment. For the most favored GPs, the reform distributed additional capitation at 8% of the total baseline income. We study the effects on the number of patients per GP and the number of services per patient, applying difference-in-difference (DiD) models. For both schemes (moderate and intensive), we find tendencies of reductions in the number of patients served and the level of service provision per patient. This also holds for complex patients indicating that the reform did not improve equity in access. The effect on income showed a 2.5% increase in the first follow-up year but the effect became insignificant in the second year after the reform. We interpret this result as a sign that GPs trade income increases with leisure as suggested by the target income hypothesis.
JEL Classification: I10, J22

Lægemiddelregistret

- DST oversight: https://www.dst.dk/extranet/forskningvariabellister/Oversigt%20over%20registre.html

- LMDB
- https://www.dst.dk/extranet/ForskningVariabellister/LMDB%20-%20L%C3%A6gemiddeldatabasen.html

- EKSD: dato
- EKSP: pris

- RECU: ydernr i SSSY